

Smart irrigation and Internet of Things

Internet of Things Smart irrigation and environmental monitoring system.

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Abstract: This chapter illustrates the process of development and creation of a smart system monitoring and automation system. founded on the Internet of Things (IoT). The system detects the temperature rate, humidity, gas in real time. Among other things, levels, and environmental conditions can be measured using sensors. The information is relayed in the IoT platform. forwarded to Wi-Fi so that you can track and make decisions remotely. The given system will make sure that the environment is more efficient and automated and as not dangerous. the alerts are registered and the information is displayed in real time. The strategy of the system, the manner in which it operates, how it Experiencing was an implementation strategy of what happened and how it could happen in. It is also in this chapter that future are discussed.

Keywords: the automation, smart monitoring, environment, and sensors. IoT.

1.Introduction:

IoT has evolved in a very fast manner and has revamped the ancient. retail systems to intelligent and automated systems. IoT enables equipments to broadcast with each other, accumulate information, and process it without the assistance of human beings. The Monitoring of the is also required to use Internet of Things (IoT). One of its most important ones is the environment. applications. One of the aspects

what can have direct. effect on health of individuals as well as business manner. measured ones consist of temperatures, humidity, concentration. of gasses and the air quality. The traditional monitoring systems are done manually and are likely to be subjected to errors. Internet of things (IoT) systems. offer:

- Real-time watching have to be very accurate: that is why this feature is offered. distantned, on fronterpoint automation This is the chapter about how to. create a smart with the use of sensors and microcontrollers. The environment can be monitored using system.

2. Survey of Literature

The IoT-based systems are popular in the agricultural sphere and environmental monitoring.

2.1 IoT for Keeping an Eye on the Environment

IoT systems consume numerous sensors that are interlinked with microcontrollers to receive information regarding the surrounding. These The systems can prove useful in: Monitoring pollution. Intelligent cities Occupational health and safety.

2.2 IoT in Agriculture (Work that is Related)

An intelligent water irrigation system. takes advantage of a few of the sensors: Sensor for soil moisture Sensor for temperature Sensor for humidity The system waters crops automatically using the IoT systems when it is. necessary. Gap in Research Small-scale There is no real time automation of systems. There are a , there is a limited number of cooperating

sensors. Required cheap and scalable solutions
Requirement of cheap and scalable. solutions
required a large number of people to utilize the
solution. Need where solutions are desired that
are can be utilized by cheap and. quite an
amount of people require. cost-effective and
scalable solutions.

3. Suggested System

3.1 Goal In order to develop a smart IoT system that:

Expects transfer of information to cloud
real- time. Sends notifications, automates
operations.

3.2. The System Organization. The system will include:

Temperature, humidity, gaseous and
other. sensors. Microcontroller (Arduino,

ESP8266 or ESP32). Internet of things Internet of
Things IOT32 EXP. Things (ThingSpeak / Blynk)

• Show (LCD / Dashboard).

4. Block Diagram:

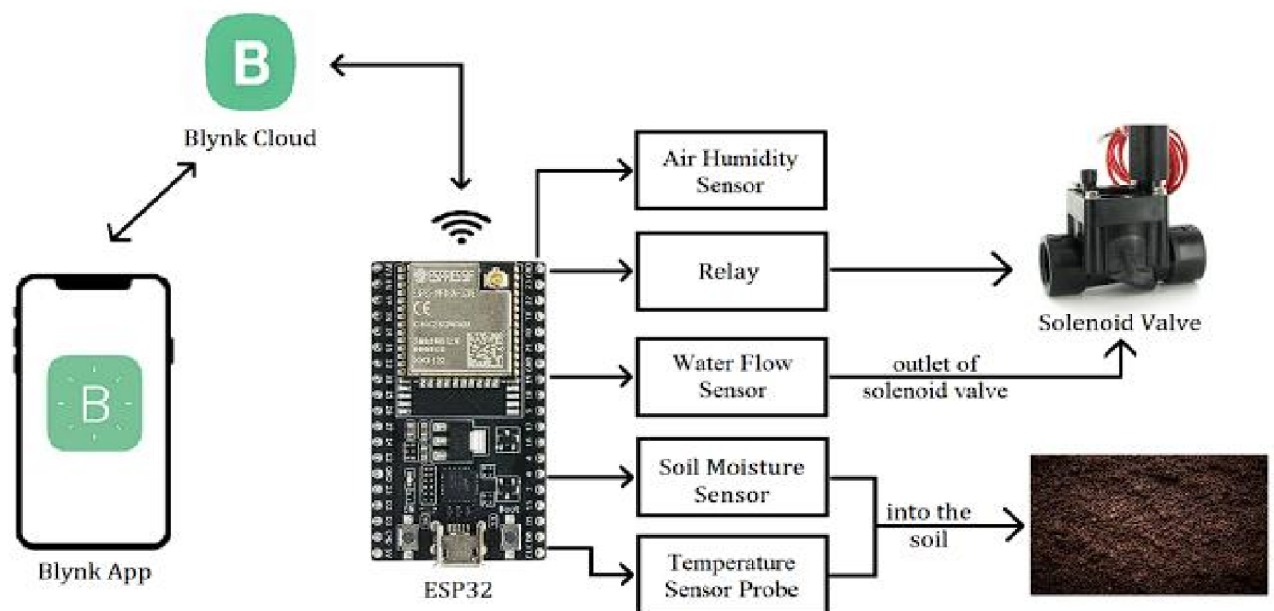


Figure 1. Overview of the IoT-enabled smart drip irrigation system [21].

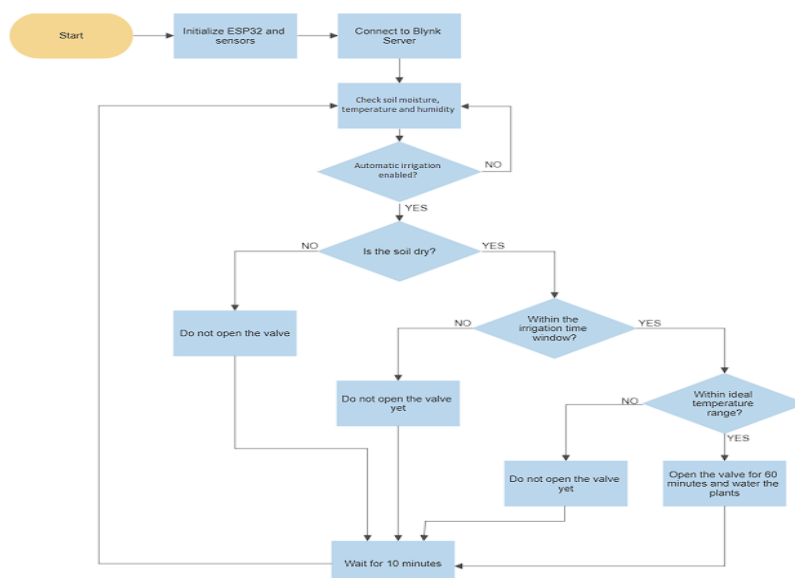


Figure 3. Flow chart of the IoT-enabled smart drip irrigation system.

5. Parts Used:

The components of the hardware will include an ESP32. Microcontroller as the controller which will be the main one to process in all. The information which the sensors will present and operate an irrigation. system. The sensor used is DHT22 which reads the presence of the level of. climate and moisture of the place. The water content of soil is determined by the use of soil. hygrometer(modeled as a potentiometer). The information received will help in making a decision as to whether to get the irrigation. it is necessary or otherwise. The LCD Display in its size. will have 16x2 role of displaying the current status. of the temperature, humidity and other information concerning the present situation with the irrigation system. Connector will be Relay Module. A switch to be utilized in turn on of the irrigation system. There is the opening and closing of the water valve in the process of. irrigation will be simulated by the Servo Motor. LED will be utilized as an indicator to demonstrate whether the irrigation process is contingent or otherwise.

6. How it works:

The system is realized as attaining the. following:

- 1.Data on sensors are recorded about. the surrounding.
2. The information is processed in the microcontroller.
3. Wi-Fi transmits to the IoT information. cloud.
4. The information is displayed in the dashboard.
5. Notifications are received in case of limits of. hit. For example: In the event that the temperature and falls over the threshold, a. notification is given. Will be shocked at the sight of gas. The system monitors and provides real time updates.

7.Implementation:

7.1 Simulation Platform Your implementation has been done on:

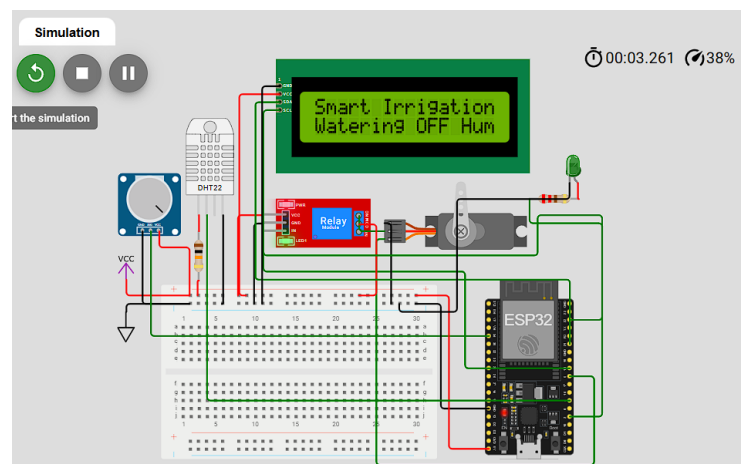
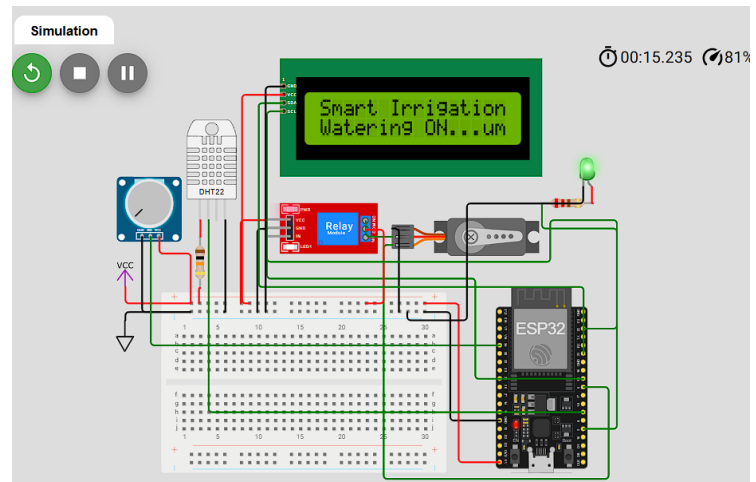
- Wokwi IoT Simulator

7.2 Features of Your Project:

- Simulation of real sensors
- Digital IoT dashboard
- Automated notifications
- Remote cloud data management

7.3 Code Explanation (Brief):

- Sensors setup Information was read in loops. Conditions applied Information communicated via Wi-Fi.



8. Results and Analysis:

The system is able to monitor irritate and water and humidity and heat. crops. The system is able to:

- Monitor environmental factors Send data to cloud in real time.
- Present data graphically Send notifications when below threshold was surpassed.

Advantages:

- Low-cost
- Precise
- Scalable
- Maintainable

We collected the irrigation data over one week and plotted the soil moisture content graph (Figure 20). The soil is dry if the moisture reading crosses 380. If the moisture reading is above 380 and the temperature and irrigation window are in the proper range, the ESP32 opens the valve to irrigate the field. The soil is considered wet if the reading is between 190 and 380, and very wet if the reading is between 0 and 190. As seen in the graph, the onions received water four times this week.

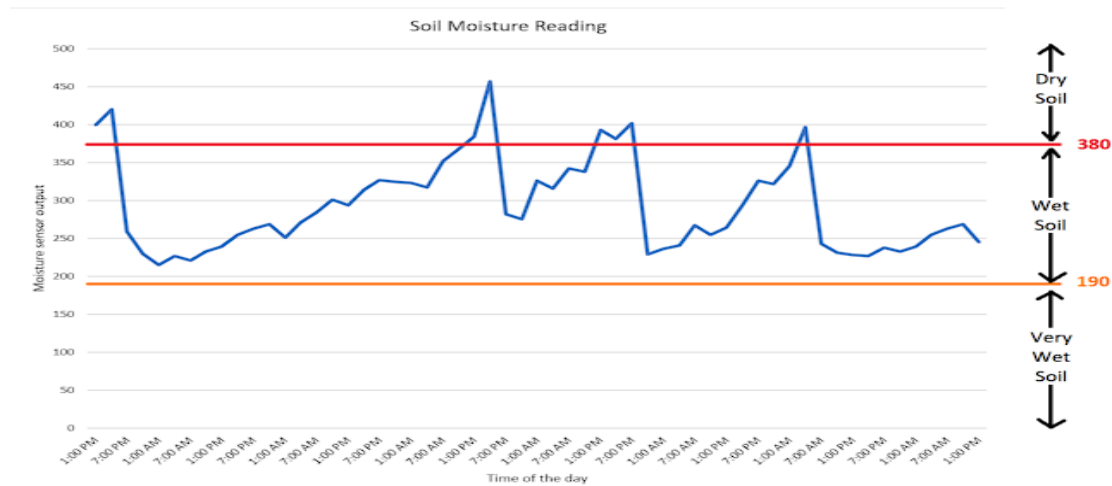


Figure 20. Soil moisture variation for one week.

Each dripper had a flow rate of 0.64 L per hour and watered an onion bulb for an hour a day for four days a week. In other words, the onion received 0.64 L of water a day. The total water provided to an onion over the week was as follows: $0.64 \times 4 = 2.56$ L (1) To convert the value into gallons, we divide Equation (1) by 3.785, as shown: $2.56/3.785 = 0.676$ gallons (2) Spring onions or shallots grown from the bulb need about an inch of water per week to grow well [44]. “One inch of water” refers to the amount of water necessary to cover one square foot of soil with one inch of water [45]. As there are 12 inches in a foot, the square inches of water needed per square foot of soil is as follows: $12 \times 12 = 144$ sq. inch (3) Moreover, 1 gallon is 231 cubic inches. Thus, when we divide 144 sq. inches by 231, we obtain the amount of water needed by onion bulbs per square foot as follows: $144/231 = 0.623$ gallons (4) As seen above, the actual amount of water provided to the onion bulbs in a week using the smart irrigation system in Equation (2) and the recommended amount of weekly water in Equation (4) are very close; hence, the smart irrigation system succeeds in providing adequate moisture to the plants. We plotted the humidity variation graph for one week, as shown in Figure 21. The humidity was mostly above the lower limit of 25%. However, corresponding to some of the hottest hours of the day, it was not uncommon for the humidity to fall well below the 25% mark.

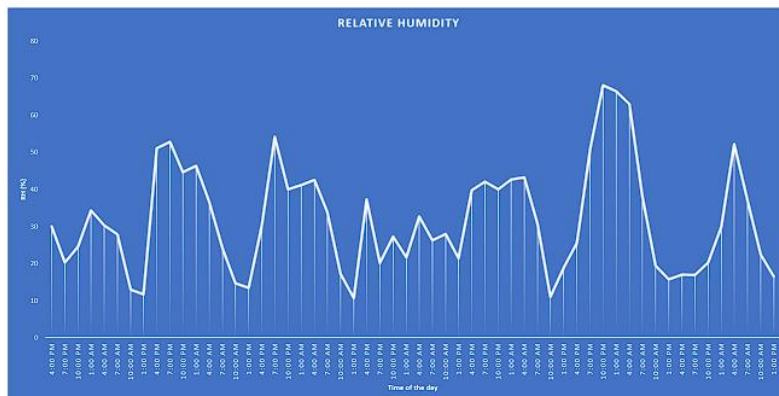


Figure 21. Humidity fluctuation throughout the week.

We plotted the temperature fluctuation graph for one week, as shown in Figure 22. The temperatures were the highest between 1 p.m. and 3 p.m. This timing also coincides with the lowest humidity readings of the day. The highest observed temperature was 46 °C. The temperature was comparatively low during the morning and evening irrigation time windows.

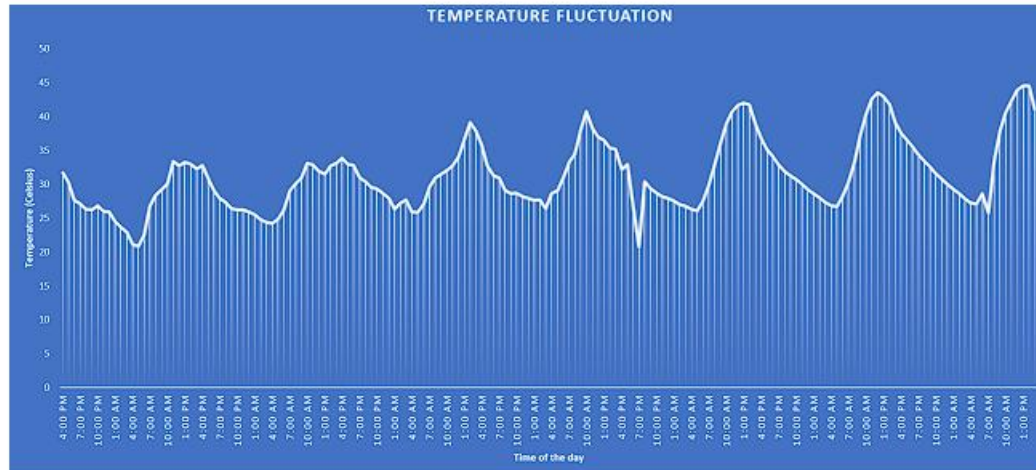


Figure 22. Temperature fluctuation throughout the week.

The moisture content, humidity, and temperature data collected are valuable for observing the weekly and monthly trends, making changes to the firmware, and are especially useful for building a greenhouse around the smart irrigation system.

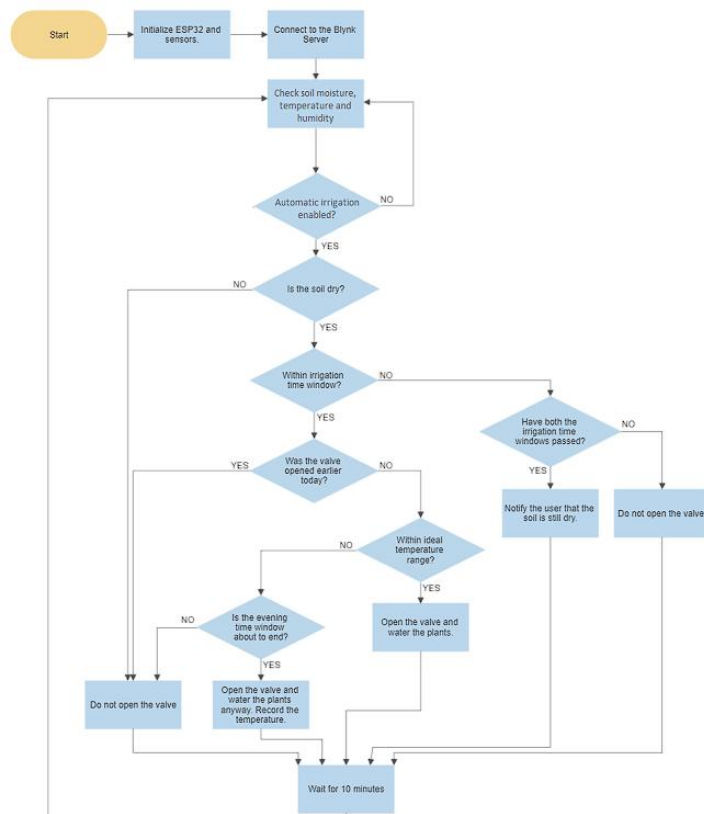


Figure 17. Final flow chart of the smart drip irrigation system firmware.

9. Applications:

- “Smart Cities,”
- Industrial Monitoring,
- Agriculture,
- Pollution Management,
- House Automation.

10. Strengths:

- Real-time monitoring,
- Remotely controlled,
- Automation,
- Energy-efficient,
- Effective.

11. Weakness:

The Internet is some way or another dependent on it.

- Inaccuracies of sensors,
- Security threats.

12. Future Scope:

Prediction based on AI technology, Mobile application incorporation,

- Solar Energy-driven System, Managerial analytics, machine learning.

13. Conclusion:

A monitoring system that observes the environment and falls outside the expertise of Coratech will monitor. IoT-based business is a very effective one. solution to the environmental monitoring. since it automates and processes data in real-time. The extension of the system is possible. unit with the assistance of AI and cloud. technologies.

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